

1. Some insight into how the interacting field theory formalism works can be obtained by studying the interaction of free fields with an external c -number source. Consider a real scalar quantum field which satisfies the equation of motion,

$$(\partial^\mu \partial_\mu + m^2)\phi(x) = j(x), \quad (1)$$

where, $j(x)$ is a classical c number source. A particular c number solution is,

$$\phi_c(x) = \int d^4y G(x-y)j(y),$$

where G is a Green's function of the operator, $(\partial^\mu \partial_\mu + m^2)$:

$$(\partial^\mu \partial_\mu + m^2)G(x-y) = \delta^4(x-y).$$

A general solution of (1) is therefore,

$$\phi(x) = \phi^0(x) + \int d^4y G(x-y)j(y), \quad (2)$$

where, $\phi^0(x)$ is a free quantum field $((\partial^\mu \partial_\mu + m_\phi^2)\phi^0(x) = 0)$. The precise form of the Green's function is specified by the boundary conditions. We assume that the current $j(x)$ is adiabatically switched on only over a finite time interval, i.e., $j(x) \rightarrow 0$, for $|t| \rightarrow \infty$. The retarded and advanced Green's functions are given by:

$$\begin{aligned} G_{ret}(x) &= -\frac{1}{(2\pi)^4} \int d^4p \frac{e^{-ip \cdot x}}{(p_0 + i\epsilon)^2 - \vec{p}^2 - m^2} \\ G_{adv}(x) &= -\frac{1}{(2\pi)^4} \int d^4p \frac{e^{-ip \cdot x}}{(p_0 - i\epsilon)^2 - \vec{p}^2 - m^2}. \end{aligned} \quad (3)$$

- (a) Perform the integral over p_0 choosing an appropriate contour and find an explicit expression for these two Green's functions which satisfy,

$G_{ret}(x) = 0$ for $x^0 < 0$ and $G_{adv}(x) = 0$ for $x^0 > 0$. For example, you should find that,

$$G_{ret}(x-x') = i \frac{\theta(x^0 - x'^0)}{(2\pi)^3} \int \frac{d^3p}{2p_0} (e^{(-ip_0(x^0-x'^0)+i\vec{p}\cdot(\vec{x}-\vec{x}'))} - e^{(ip_0(x^0-x'^0)+i\vec{p}\cdot(\vec{x}-\vec{x}'))}), \quad (4)$$

with a similar expression for $G_{adv}(x-x')$. Recall that $p_0 = \sqrt{(\vec{p})^2 + m^2}$

(b) Show that,

$$G_{ret} - G_{adv} = \frac{i}{(2\pi)^3} \int \frac{d^3p}{2p_0} (e^{(-ip_0(x^0-x'^0)+i\vec{p}\cdot(\vec{x}-\vec{x}'))} - e^{(ip_0(x^0-x'^0)+i\vec{p}\cdot(\vec{x}-\vec{x}'))}), \quad (5)$$

and relate it to the commutator function $[\phi_{in}(x), \phi_{in}(y)]$ which was calculated in class.

Thus, we may write (5), taking into account the boundary conditions, as

$$\begin{aligned} \phi(x) &= \phi_{in}(x) + \int d^4y G_{ret}(x-y) j(y) \\ \phi(x) &= \phi_{out}(x) + \int d^4y G_{adv}(x-y) j(y). \end{aligned} \quad (6)$$

The fields, ϕ_{in} and ϕ_{out} describe the scalar field before and after interaction with the source $j(x)$. Thus in a formal sense we have,

$$\begin{aligned} \lim_{x^0 \rightarrow -\infty} \phi(x) &= \phi_{in}(x) \\ \lim_{x^0 \rightarrow \infty} \phi(x) &= \phi_{out}(x). \end{aligned} \quad (7)$$

The precise mathematical meaning requires taking a "weak" limit in the sense that some local average of the fields between normalized states satisfies these relations. This limit crucially requires the assumption of the adiabatic vanishing of the source at $|t| \rightarrow \infty$. We have seen in class that the S matrix connects the "in" and "out" fields:

$$\phi_{out} = S^{-1} \phi_{in}(x) S. \quad (8)$$

We will next explicitly construct the S matrix for this problem. Indeed from (8) it follows that

$$\phi_{out}(x) = \phi_{in}(x) + \int d^4y (G_{ret}(x-y) - G_{adv}(x-y)) j(y). \quad (9)$$

Using the result of part (b) above, we therefore have,

$$\phi_{out} = S^{-1} \phi_{in}(x) S = \phi_{in}(x) + i \int d^4y [\phi_{in}(x), \phi_{in}(y)] j(y) \quad (10)$$

(c) From Eq. (12) explicitly determine the S matrix upto an unobservable c number phase.